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Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques

Lecture Notes - 10

Topics covered :

1. First Order Autonomous Systems
2. Equilibrium points and Stability

- In this lecture and the related analysis that follows, we will focus on first-order ordinary differential equations (ODEs), where time is the independent variable and the dependent variable is denoted as y. The term Trajectory refers to how the solution of the differential equation changes over time.

- Differential equations in which the independent variable does not appear explicitly are called autonomous differential equations. Mathematically, autonomous differential equations are of the form

$$\frac{dy}{dt} = f(y) \tag{1}$$

where $f(y)$ can be either linear or nonlinear. For example, the current response of an RL circuit driven by a DC voltage source is described by the equation

$$Ri(t) + L\frac{di}{dt} = V_{dc} \implies \frac{dy}{dt} = \frac{V_{dc} - Ri(t)}{L} \tag{2}$$

This is an example of an autonomous differential equation.

- Example: Exponential and Logistic growth: [1,2] Consider $y = y(t)$ as the population of a given species at time t. The simplest hypothesis regarding population variation assumes that the rate of change of y is proportional to its current value. This leads to the model of exponential growth i.e.,

$$\frac{dy}{dt} = ry, \quad y(0) = y_0 \qquad \implies \qquad y = y_0 e^{rt}. \tag{3}$$

Here, the constant r is the growth (or decline) rate, depending on whether $r > 0$ (growth) or $r < 0$ (decline). Intuitively, r represents the net difference between birth and death rates. If $r > 0$, the model predicts exponential population growth over time, as shown for different initial values of y_0. However, this model suggests that the population grows indefinitely, which is unrealistic. To address this, we use the logistic population model, which accounts for environmental limitations such as space, food supply, or other resources. This model assumes that the growth rate $h(y)$ depends on the current population i.e.,

$$\frac{dy}{dt} = h(y) \cdot y. \tag{4}$$

Here, the simplest form of $h(y)$ is:

$$h(y) = r\left(1 - \frac{y}{K}\right), \tag{5}$$

where r is the intrinsic growth rate, representing growth in the absence of limiting factors, and K is the carrying capacity of the environment (i.e., the maximum population the environment can support). Substituting $h(y)$ into the differential equation, we obtain the logistic equation (also known as the Verhulst equation)

$$\frac{dy}{dt} = r\left(1 - \frac{y}{K}\right) \cdot y. \tag{6}$$

This equation is nonlinear but autonomous, offering a more realistic model of population growth by incorporating the effect of limiting resources.

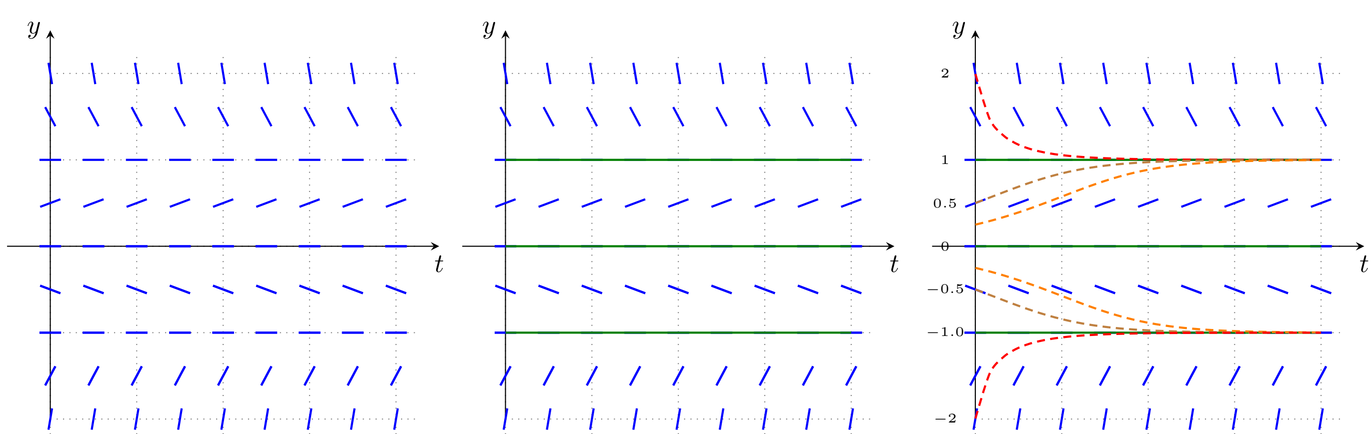
- Autonomous differential equations are separable, meaning an analytical solution can be obtained by integrating both sides. However, in some cases, this integration may be difficult or even impossible to evaluate. Even otherwise, it is possible to analyze the solution of an autonomous differential equation using graphical techniques, without explicitly solving for the solution.

- One important concept when analyzing autonomous differential equations graphically is that of equilibrium points (or critical points). An equilibrium point $y = y^*$ of an autonomous differential equation is a solution where $f(y^*) = 0$, meaning the rate of change of the solution is zero. In other words, the solution does not change over time at the equilibrium point. Trajectories that start at an equilibrium point will remain at that point for all future time, and these solutions are often referred to as steady-state solutions.

  Consider the autonomous DE
  $$\frac{dy}{dt} = y - y^3 \tag{7}$$

  The equilibrium points of the system, determined by solving $f(y) = 0$, are $y^* = 0$ and $y^* = \pm 1$. The slope fields corresponding to this autonomous DE are shown in Fig. 1(a).

  It is important to note (from Fig. 1(a)) that the slope fields generally indicate a direction towards one of the equilibrium points, though not necessarily the nearest one. In Fig. 1(b), the solution with an initial condition at the equilibrium point $y = y^*$ is shown. As expected, this leads to the solution remaining constant along the



(a) Slope fields (b) Trajectories with $y_0 = y^*$ (c) Trajectories

Fig. 1: Slope fields and trajectories of the autonomous ODE $y' = y - y^3$

line $y = y^*$.

By analyzing the slope fields (Fig. 1(a)), we can predict the behavior of the solution over time for initial conditions that are not at the equilibrium points. For example, when the initial condition is $y_0 = 2$, the direction field suggests that the trajectory will approach the equilibrium point $y^* = 1$. This is also the case for the initial condition $y_0 = 0.25$, as shown in Fig. 1(c). Conversely, when $y_0 = -2$, the trajectory tends towards the equilibrium $y^* = -1$. Interestingly, even when the initial condition is close to $y^* = 0$, the trajectory moves towards the other equilibrium points, not towards $y^* = 0$.

The analysis of how trajectories evolve over time for initial conditions other than the equilibrium points relies on information from the slope fields. These fields convey two key pieces of information: (a) the location of the equilibrium points, and (b) the slope of the solution at any given point, which is determined by the function $f(y)$ (since the slope is equal to $f(y)$).

A more intuitive and useful graphical tool for analyzing the long-term behavior of solutions is the y-$f(y)$ plane (or equivalently, a graph with the y-axis as $f(y)$ and the x-axis as y). In higher dimensions, this type of representation is referred to as phase portraits.

In this representation, the equilibrium points are identified and marked. For each initial point other than the equilibrium, the direction of the trajectory is indicated using arrows. Since y is plotted along the x-axis, the arrows are placed on this axis to represent the flow of the system. Typically, there are regions of the phase space where trajectories all move in the same direction, so it is not necessary to mark the trajectory for every possible initial condition. Most importantly, the direction in which the trajectories move is determined by the sign of $f(y)$. If $f(y) > 0$ (i.e., $y' > 0$), the solution increases over time, and the arrows point to the right. If $f(y) < 0$ (i.e., $y' < 0$), the solution decreases, and the arrows point to the left. Since the system involves smooth functions, the solutions typically move towards one of the equilibrium points or infinity. An equivalent phase portrait for the ODE $y' = y - y^3$ is shown in Fig. 2.

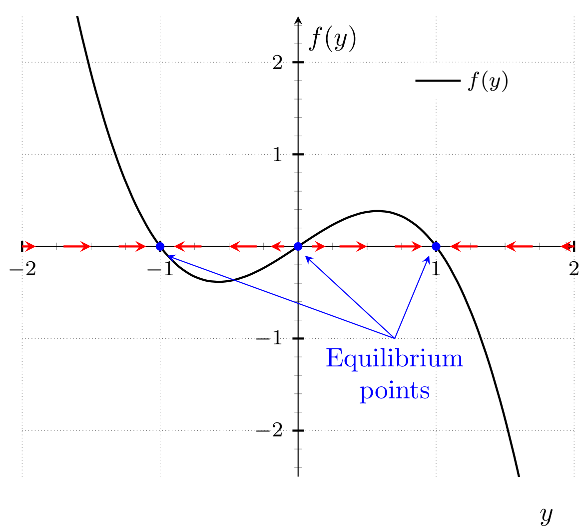


Fig. 2: Phase Portrait of autonomous ODE $y' = y - y^3$

- Example: Construct the phase portrait corresponding to the current response of an RL circuit with DC source with $R = 1\Omega$, $L = 1$ H and $V_{dc} = 2$.

The autonomous DE corresponding to the given circuit is

$$\frac{di}{dt} = (2 - i) \tag{8}$$

The DE has one equilibrium point namely $i = 2$ and the trajectory at points other than $i = 2$ converges towards 2. The phase portrait of the circuit is shown in Fig. 3.

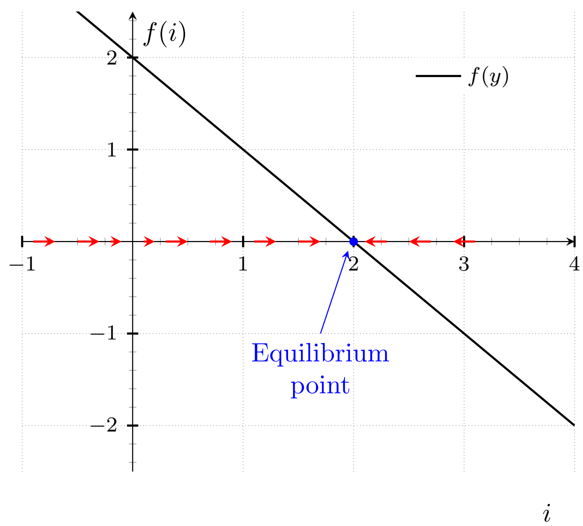


Fig. 3: Phase Portrait of autonomous ODE $i' = 2 - i$

- **Plotting Phase portraits of first-order Autonomous ODE**:

  1. *Equilibrium points:* Determine the value/s of y where $f(y) = 0$ and mark the equilibrium points
  2. *Determine how the trajectory evolves:* If $f(y) > 0$, the trajectory moves to the right (increasing in y). If $f(y) < 0$, the trajectory moves to the left (decreasing in y).
  3. *Draw arrows for trajectory evolution:* Draw arrows pointing to the right for regions where $f(y) > 0$. Draw arrows pointing to the left for regions where $f(y) < 0$.

- **Introduction to the stability of Equilibrium points [3]**: An analysis of the phase portrait for the differential equation $y' = y - y^3$ (shown in Fig. 2) reveals that for initial points near the equilibrium points $y = -1$ and $y = 1$, the solution approaches these equilibrium points as $t \to \infty$. A similar behavior is observed for the equilibrium point $i^* = 2$ in the autonomous ODE $i' = 2 - i$ (shown in Fig. 3). However, a key difference is that, regardless of the initial value, the solution always approaches the equilibrium point $i^* = 2$.

  The equilibrium points $y = +1$ and $y = -1$ for the ODE $y' = y - y^3$ are locally stable equilibrium points. This means that if the initial value y_0 is close to either of these equilibrium points, the solution will approach the equilibrium as $t \to \infty$.

  In contrast, the equilibrium point $i^* = 2$ for the autonomous ODE $i' = 2 - i$ is a globally stable equilibrium point. This implies that, no matter the initial value i_0, the solution will always approach the equilibrium point as $t \to \infty$.

On the other hand, the equilibrium point $y = 0$ for the ODE $y' = y - y^3$ is unstable. This means that even when the initial point is close to $y = 0$, the solution will diverge away from the equilibrium as $t \to \infty$.

- Formally, an equilibrium point y^* is locally stable if $\forall \epsilon > 0$, $\exists \delta > 0$ such that if $|y_0 - y^*| < \delta$, then $|y(t) - y^*| < \epsilon \ \forall t \geq 0$. In simple terms, an equilibrium point y^* is locally stable if, for any small perturbation of the system from y^*, the solution returns to y^* as time goes to infinity.

- An equilibrium point y^* is globally stable if all trajectories, irrespective of the initial condition move towards the equilibrium point as $t \to \infty$.

- An equilibrium point y^* is unstable if for small perturbation, the trajectories move away from the equilibrium point as $t \to \infty$.

- **Linearization and Local Stability**: In many real-world applications, it is crucial to analyze the behavior of nonlinear autonomous ODEs, particularly around equilibrium points. A key aspect of this analysis is determining whether the equilibrium point is locally stable. This type of analysis is typically carried out by linearizing the system around the equilibrium point and examining its stability.

  Let y^* represent the equilibrium point about which we wish to perform the analysis and linearization. Let $\Delta y(t)$ denote the deviation of the solution $y(t)$ from the equilibrium y^*, so that the solution can be expressed as $y(t) = y^* + \Delta y$. The value of $\Delta y(t)$ at $t = 0$ corresponds to a small perturbation from the equilibrium point. In this context, the equilibrium point is considered locally stable if, for small perturbations in the initial condition, the solution $y(t)$ approaches y^* as $t \to \infty$. Another way to frame this is that $\Delta y(0)$ represents a small deviation of the initial condition from the equilibrium, and the analysis of local stability determines whether the system will return to the equilibrium point y^* over time.

  To study the local stability of the nonlinear autonomous ODE

$$\frac{dy}{dt} = f(y) \tag{9}$$

  we linearize the system around the equilibrium point y^*, where we wish to assess the stability. Using Taylor's series expansion, we approximate $f(y)$ around y^* i.e.,

$$f\left(y^* + \Delta y\right) \approx f\left(y^*\right) + \Delta y f'\left(y^*\right) \approx \Delta y f'\left(y^*\right) \text{ since } f\left(y^*\right) = 0 \tag{10}$$

  On the other hand, since y^* is an equilibrium point, its time derivative is zero. Therefore, the total derivative of $y(t)$ is simply

$$\frac{dy}{dt} = \frac{d}{dt}\left(y^* + \Delta y\right) = \frac{d\Delta y}{dt} \text{ since } \frac{dy^*}{dt} = 0 \tag{11}$$

  Thus, the linearized system around the equilibrium point is given by

$$\frac{d\Delta y}{dt} = \lambda \Delta y \text{ where } \lambda = f'\left(y^*\right) \tag{12}$$

1. If $\lambda = f'(y^*) < 0$, $\Delta y \to 0$ as $t \to \infty$ indicating that $y(t) \to y^*$. This implies that the equilibrium point is locally stable.

2. On the other hand, if $\lambda = f'(y^*) > 0$, $\Delta y \to \infty$ as $t \to \infty$ indicating that $y(t)$ moves away from y^*. This implies that the equilibrium point is unstable.

References

[1] W. E. Boyce, R. C. DiPrima, and D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th ed. Hoboken, NJ: Wiley, 1997.

[2] C. H. Edwards and D. E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th ed. Upper Saddle River, NJ: Pearson, 2000.

[3] S. H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd ed. Cambridge, MA: Perseus Books, 2018.